Transport and Fitness for purpose: assessment, baseline benchmark and steps to define an IMR plan for new CO2 pipeline and for repurposing.

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1 ABSTRACT

In the last 20-25 years the scientific community has investigated many aspects of transporting CO_2 for reinjection and storage from CO_2 specific flow assurance issues to material compatibility starting from pure to anthropogenic compositions. The operating scenarios include the design of new pipelines and the repurposing of existing assets (when deemed feasible).

The paper presents an updated view of the state of art of the scientific knowledge on the fluid being transported and proposes a step-by-step guidance for developing an inspection, maintenance and repair plan considering both the conversion of existing offshore pipelines from hydrocarbons to CO_2 transport and reinjection, and the design of brand new asset for CO_2 transport. In particular, the main focus of the study will be the analysis of the impact of the new transport conditions and new fluid characteristics, on threats that still needs to be considered, as well as the investigation of new threats arising in the new transport scenario.

The paper discusses all the activities required to start the initial assessment and to reach the pipeline baseline benchmark, for both repurposing of existing pipelines and design of new pipelines for CO_2 supply. Based on this, it is possible to start detailed risk assessment activities that are presented for a real case. The feasibility of the plan is discussed in terms of the challenges related to the fluid being transported, the inspection technologies available and the maintenance and repair strategy to be put in place to guarantee the system functionality in normal conditions and in case of relevant damage to the assets.

2 INTRODUCTION

The climate crisis, one of the biggest challenges of the 21st century, and the consequent urgent need to capture and transport anthropogenic CO_2 to permanent storage or reuse facilities, has led to many new industrial scenarios. Since the 1970s, CO_2 transport for Enhanced Oil Recovery has been optimized for dense phase transport, balancing compression investment with improved reservoir recovery.

According to the IEA CCUS Projects Database (updated in 2024), only six projects were operational between 1972 and 2000 with a clear emissions reduction scope. Today, there are 44 projects in operation, with two having the target to exceed 10 Mt CO_2/year (Alberta Carbon Trunk Line and Petrobras Santos Basin pre-salt oilfield CCS).

Current and future industrial scenarios involve capturing impure CO_2 from various sources, including gaseous CO_2 transport and the assessment of the impact of associated impurities. International standardization bodies like ASME, ISO, and DNV are working to supplement the new scenarios design and operating requirements.

Purpose of the paper is to provide a brief state of the art of knowledge and standardization for the design of new or repurposed pipelines offshore for CO_2 transport, then to enter more specifically in the field of the design related to the pipeline safe operation and maintenance analyzing how the requirements may differ with respect to conventional O&G pipelines. The analysis involves the application of the most reliable methodologies specified for the IMR planning and execution.

3 TRANSPORTING CO_2 - STATE OF ART KNOWLEDGE

The behavior of carbon dioxide (CO_2) pipeline systems differs significantly from traditional hydrocarbon pipelines, especially when CO_2 is transported in liquid/dense phase and it is influenced by impurities and free water. Here are some key differences:

1. **Phase Behavior**: CO_2 can exist in gas, liquid or supercritical (dense) phase. In dense phase, CO_2 behaves differently compared to hydrocarbons, requiring specific design considerations.
2. **Impurities**: The presence of impurities like hydrogen sulfide (H_2S), nitrogen (N_2), and water can significantly affect the phase behavior and corrosion rates of CO_2 pipelines. Moreover, impurities can lower the solubility limit of water in CO_2, leading to potential corrosion issues.
3. **Corrosion**: CO_2 pipelines are more susceptible to corrosion, especially in the presence of free water. This necessitates the use of corrosion – resistant materials or the implementation of a stringent drying process to ensure the CO_2 stream is free of water.
4. **Safety and Monitoring:** due to the unique properties of CO_2, pipelines safety systems, including overpressure protection and leak detection systems, may require specialized design/devices to ensure safe operations.

These differences highlight the need for tailored engineering solutions and operational procedures to safely and efficiently transport CO_2 through pipelines. The limited industry understanding, testing and operational data indeed create gaps in the existing standards and guidelines. This can hinder accurate assessment of design requirements and lead to overly conservative approaches. To address these gaps, ongoing research and development are crucial.

3.1 Phase Behavior

CO_2 pipelines are engineered to function above the critical pressure and temperature, ensuring the fluid remains in a single-phase during transportation. Maintaining CO_2 in a single-phase during pipeline transport is advantageous as it guarantees stable flow, predictable pressure drops and efficient transport with higher capacity. This method also reduces risks of corrosion, erosion, and operational issues such as slugging while simplifying pipeline design and minimizing costs. Conversely, two-phase transport introduces complexities like unstable flow, higher energy demands, and increased safety risks, making single-phase transport the preferred and more reliable option.

Understanding the critical pressure and temperature of pipeline fluids is essential for effective operation. At temperatures and pressures exceeding the critical values (31.1°C and 73.8 bar for pure CO_2), CO_2 fluid transits into a supercritical phase that combines gas-like diffusion and liquid-like density.

3.2 CO_2 Specification

The impurities that require stringent control and monitoring in the CO_2 specification are as follows:

- Constituents that influence the phase envelope and can cause CO_2 corrosion (e.g. methanol and glycol)
- Constituents that enhance degradation or cracking
- Impurities such as H_2O, O_2, SOX, NOX, H_2S which reacting with each other can give rise to very strong acids
- Other constituents for which the impact is not fully determined but need control.

During the operating phase, the CO_2 composition with respect to critical impurities need to be monitored. Impurities in the CO_2 stream can affect its thermophysical properties, which in turn can affect the efficiency and cost of CCS technology. The presence of impurities can affect the <u>phase behavior</u> altering the gas-liquid equilibrium, the <u>density</u> that is vital for designing pipelines, the <u>speed of sound</u> that is important for designing pipeline to prevent running ductile fracture, the <u>viscosity</u> necessary for pressure drop calculation.

3.3 Water Solubility

Here below some critical points regarding the challenges of handling CO_2 in presence of free water, in carbon steel pipelines:

- **Corrosion and Fatigue**: Free water combined with CO_2 and related impurities can lead to significant corrosion and fatigue issues in carbon steel pipelines. This is a major concern for the integrity and longevity of the infrastructure.
- **Flow Assurance and Hydrates**: The presence of water can also lead to flow assurance problems, including the formation of hydrates, which can block pipelines and disrupt operations.
- **Water Solubility in CO_2**: CO_2 has a high solubility for water, particularly in its dense phase, which helps in dewatering. However, this solubility is highly sensitive to impurities, temperature and pressure.
- **Risk of Water Dropout**: Changes in phase (e.g., venting), mixing CO_2 streams of different compositions, or significant temperature/pressure changes can reduce CO_2's water solubility, leading to water dropout. This can exacerbate corrosion and other issues.
- **Solubility Behavior**: In the vapor state, CO_2's ability to dissolve water increases with higher temperatures and lower pressures, similar to natural gas. However, when transitioning to the liquid state, solubility increases with pressure, which is the opposite of what happens in the vapor state.

Understanding these factors is crucial for designing and operating CCS systems effectively.

3.3.1 Hydrate Formation

The formation of hydrates is influenced by pressure, temperature, water concentration and impurities in the CO_2 stream. This needs to be considered for all operating conditions, both onshore and offshore. Typically, hydrate formation is not a concern under normal operating conditions but can become problematic at low temperatures if the CO_2 is not sufficiently dewatered.

The presence of impurities, such as hydrogen, nitrogen and carbon monoxide, can shift the temperature at which hydrates form to higher levels.

3.4 Industry Standards & Guidelines

For the safe management of pipeline infrastructure, industry standards, recommended practices, and guidelines play a crucial role. These standards ensure that materials, products, processes and services meet the required quality, reflecting industry experience and fostering trust and confidence among stakeholders, authorities and society.

For pipeline transportation of CO_2, there are existing design codes that provides design criteria and guidance specifically for CO_2 transportation such as e.g. DNV-RP-F104 and ISO 27913. Figure 1 provides a high-level overview of existing codes across the CCS value chain within the ISO and DNV regimes and other Guidelines currently available.

Figure 1 Summary of standards applicable to Offshore CO_2 Pipelines

DNV	International Standard	Other
DNV – ST – F101 Submarine Pipeline Systems	**ISO 27913** Carbon dioxide capture, transportation and geological storage — Pipeline transportation systems	**AMPP GUIDE 21532** Guideline for Materials Selection and Corrosion Control for CO2 Transport and Injection
DNV – RP – F 104 Design and Operation of carbon dioxide pipelines	**ISO/TR 27925** Carbon dioxide capture, transportation and geological storage - Cross cutting issues - Flow assurance	**Energy Institute** Repurposing and Design Guidelines for Carbon Dioxide Pipeline
DNV-SE-0657 Re-qualification of pipeline systems for transport of hydrogen and carbon dioxide	**ISO/TR 27921** Carbon dioxide capture, transportation, and geological storage — Cross Cutting Issues - CO_2 stream composition	
DNV-RP-F116 Integrity Management of submarine pipeline systems	**ISO 13623** Petroleum and natural gas industries - Pipeline transportation systems *AMENDMENT 1: Complementary requirements for the transportation of fluids containing carbon dioxide or hydrogen*	

4 INTEGRITY MANAGEMENT PLAN – METHODOLOGY

An Integrity Management Program is designed to ensure the continual integrity of a pipeline system through comprehensive risk assessments and structured planning of various activities. The primary steps include threat identification, risk assessments, and the development of both long-term and short-term plans. These steps guide the execution of inspections, monitoring, testing, integrity assessments, and necessary repairs.

The program prioritizes understanding and mitigating risks to maintain the safety of the pipeline throughout its life cycle. By employing a risk-based approach, integrity management activities are prioritized and scheduled based on their capacity to measure and manage threats effectively, ensuring that associated risks are maintained within acceptable limits. The outcomes of the risk assessments inform the development of detailed long-term integrity management programs that specify the activities to be conducted and their respective timelines.

DNV-RP-F116 is the reference standard for the integrity management of submarine pipeline systems and will be used for the scope of the present paper: it is generally applicable to offshore pipelines regardless of content and can therefore also be applied to carbon dioxide pipelines.

The basic steps to be followed when performing a risk assessment can be resumed as follows:

- <u>**THREAT IDENTIFICATION**</u>
  Identification of all credible threats and failure modes threatening the pipeline system (Ref. 2, Table 4-1).
- <u>**PROBABILITY OF FAILURE (PoF)**</u>
  Estimate the likelihood of the identified threats, categorized into 5 PoF levels (Ref. 2, Table F-7).
- <u>**CONSEQUENCE OF FAILURE (CoF)**</u>
  Assess the consequences of each identified threat, considering factors the content of the pipeline, the internal conditions, the failure mode and the physical location (associated with factors like population, water depth, environmental sensitive area etc.), see Ref. 2 (Table F-10 to Table F-15).
- <u>**RISK LEVEL**</u>
  Combines PoF and CoF to estimate the risk level (and the inspection frequency) using a risk matrix specific to the system, which determines inspection frequency intervals (Ref. 2, Table F-1 and Table F-3).

DNV-RP-F116 guideline provides various methods for evaluating PoF and CoF, requiring different levels of effort and accuracy.

5 THREAT ANALYSIS IN CO_2 ENVIRONMENT

For a pipeline system transporting CO_2, the evaluation of threats must be carried out considering both the specific transport conditions associated to the fluid and the transport environment created by the presence of the fluid itself. This allows to:

- Understand if and how the common pipeline threats (Ref. 2, Table 4-1) can be affected by the presence of CO_2,
- Identify pipeline threats specific to CO_2 transport and those shared with other submarine pipelines.

In this way, potential hazards on a pipeline can be reviewed, putting particular emphasis on the impact of carbon dioxide on the evaluation of the probability of failure (PoF), considered more as a 'threat-related' quantity. The review of potential hazards in a pipeline system transporting CO_2 highlights that, while some threats did not evidence changes from hydrocarbon to CO_2 service (i.e. the approach to be used does not require further considerations), an in-depth analysis is required for other specific threats.

Internal Corrosion
According to Ahmad Amirhilmi A. Razak et al. (Ref. 3), the presence of water or other impurities in the CO_2 stream could activate several corrosion mechanisms and affect carbon dioxide physical properties in a different way with respect to what is normally observed in pipelines transporting hydrocarbons. Sulphur oxides (SOx) and hydrogen sulphide (H_2S) have the ability to turn sour on its own or by reacting with water to produce corrosive sulphuric acid (H_2SO_4) (Ref 3). Impurities such as SO_2, NO_2 and O_2 have the tendency to reduce the solubility limit of water (resulting in the dropout of water from the gas stream and so producing carbonic acid) and therefore giving rise to corrosion.
SO_2 and N_2 have also the potential to form sulphuric and nitric acid in presence of water, promoting an aggressively corrosive environment and metal loss (see Ref 3).
Finally, it is noted that impurities such as Oxygen, SO_2, NO_2 and H_2S can also affect Corrosion Resistance Alloy (CRA).

Ductile Fracture Propagation
Compared to natural gas pipelines a running ductile fracture is of bigger concern for carbon dioxide pipelines. In case of a pipeline fracture the fluid decompresses, but when the CO_2 reaches saturated conditions the decompression speed can become lower than the speed of a running fracture, thus challenging. Based on this, a dedicated analysis of this threat is recommended.
Although not addressed inside DNV-RP-F116 (Ref. 2), it is common that O&G Operators' Risk Analysis Methodologies for Offshore/Onshore Pipelines (RBI Development) include the ductile fracture propagation inside the structural threats group (Ref.2, Table 4-1).

External Coating Damage
If the external coating is damaged, the pipeline protection could fail giving rise to external corrosion mechanism. Normally, external coating damage can be caused by the interaction with third party (e.g., vessel, fishing gear, dropped objects for offshore pipelines, excavators for onshore pipelines, impact with vessel in case of platform risers etc.) or other reasons (installation errors), but for pipeline transporting CO_2, it must be considered that an incidental or uncontrolled depressurization of the pipeline may cause such lower temperatures (if compared to traditional oil/gas pipelines) to

potentially have an impact on external coating integrity (see Ref. 1 and Ref. 6), including potential effect on the cathodic protection.

Non-Metallic Materials

Unlike hydrocarbons transportation, the presence of non- metallic materials give rise to potential new threats for the pipeline system. In particular, the impact of CO_2 on non-metallic materials shall be considered in terms of explosive decompression of absorbed CO_2 and deterioration of mechanical properties.

When internal coating is meant to act inside a CO_2 environment, the risk of detachment from the base pipe material in a potential low temperature condition associated with rapid pipeline depressurization should be considered (Ref. 1).

The suitability of polymers and elastomers for exposure to CO_2 and impurities in dense phase CO_2 service needs to be considered and various failure modes of polymeric materials have been identified, such as (1) Reduction of (desirable) properties such as strength, stiffness, and resistance to permeation, (2) Swelling of the elastomers (attributed to the solubility/diffusion of the CO_2 into the bulk material), (3) Explosive decompression attributed to the gases that have permeated into the bulk material, and (4) Low-temperature flexibility resulting in reduced or failed sealing (Ref. 5 and Ref. 6).

Finally, threat assessment for pipeline systems transporting CO_2 requires some additional considerations (Ref. 5):

- Third Party Threats: no impact is observed on third party threats if CO_2 transportation is considered, on the assumption that an adequate level of protection is guaranteed with reference to CO_2 specific physical properties.
- Structural threats: in case of hydrocarbon pipeline re-purposed for CO_2 transport, intervention works designed and built for pipeline stabilizations (i.e. rockdump) must be carefully re-assessed: the higher density of CO_2 could represent a threat for the stability of the intervention works (e.g. slumping of rockdumps). The verification of IWs stability could require a periodical monitoring by means of external survey.
- Incorrect Operations: according to Ref. 2, threats related to incorrect operations are normally covered by review/audits and training of personnel. However, events like incorrect flow conditions (bi-phase flow) and CO_2 behaviour in case of decompression (low temperatures) can have an important impact on the management of other threats. This aspect should be captured during the risk assessment session.

6 Inspection Plan

The present section provides a description of the approach to be used to develop an inspection and monitoring program for a pipeline system transporting CO_2.

The evaluation of the Probability of Failure is carried out considering the Level-1 assessment (Flow Chart method) as per Ref. 2 : it allows to evaluate the impact of CO_2 environment in terms of new threats and ‘classic’ threats in a simple but efficient and rational way at the same time.

6.1 Internal Corrosion

An adjustment of the Flow Chart for the determination of the Probability of Failure for the internal corrosion threats is here proposed with the aim to take into account the operation inside CO_2 environment (see Figure 1 and Table 1).

Monitoring of internal corrosion threats is carried out by performing regular pipeline internal inspection by means of intelligent tool running inside the pipeline.

Figure 2: Flow chart internal corrosion gas export – DNV-RP F116 [2]

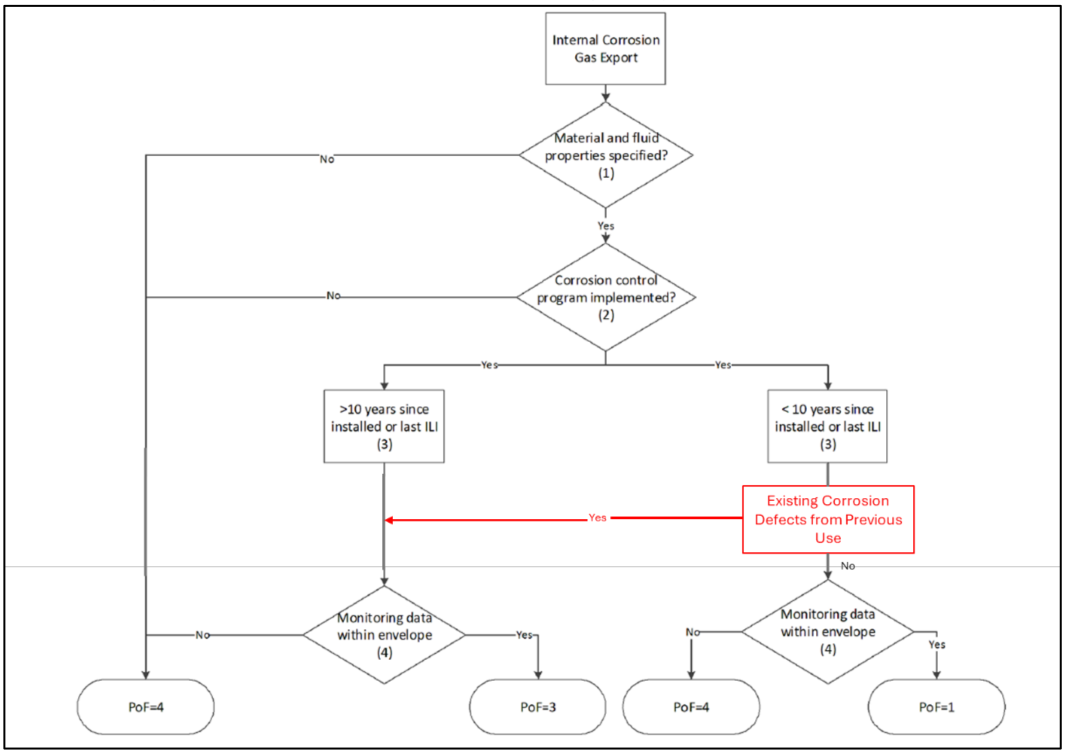


Table 1: Internal Corrosion Assessment – DNV-RP F116 [2]

ID	Question	Note	Relevant information for PoF assessment: HYDROCARBON	Relevant information for PoF assessment: CARBON DIOXIDE NEW DESIGN	Relevant information for PoF assessment: CARBON DIOXIDE RE-PURPOSING
1	*Material and fluid properties specified?*	The level of impurities that can activate internal corrosion shall be determined for every corrosion mechanism and kept within the acceptable limits together with the maximum water content.	Carbon steel Corrosion allowance Maximum water content (alt. water dew point), CO_2 and H_2S content, P, T	Carbon steel Corrosion allowance Maximum water content (alt. water dew point), Sulphur Oxides (SOx) and Hydrogen Sulphide (H_2S), Sulphur Dioxide (SO_2), Nitrogen Dioxide (N_2) and Oxygen (O_2), P, T	
2	*Corrosion control program in place?*	A product monitoring system adequate for the monitoring of impurities that can trigger internal corrosion inside a CO_2 environment must be in place.	Product monitoring (e.g. water content, H_2S) Corrosion monitoring (corrosion probes) Operational parameter monitoring (P, T)	Product monitoring: Sulphur Oxides (SOx) and Hydrogen Sulphide (H_2S), Sulphur Dioxide (SO_2), Nitrogen Dioxide (N_2) and Oxygen (O_2). Corrosion monitoring (corrosion probes) Operational parameter monitoring (P, T)	
3	*Time since installation or last ILI*	Frequency of inspection to detect corrosion defects shall be defined and respected.	Confirmation of adequate corrosion control. If time since installation or last ILI is more than 10 years condition assessment should be carried out.		
3.a	*Existing Corrosion Defects?*	Re-purposing: latest inspection data should be available in order to evaluate the internal condition of the pipeline and its suitability for future service as a carbon dioxide pipeline. In case of existing corrosion defects PoF ≥ 3 is suggested (eve if last in-line inspection < 10 years (see Figure 1).	-	-	Fitness for Service assessment of existing corrosion defects according to Industry recognized standard
4	*Monitored data within envelope and explicitly evaluated and documented on a regular basis?*	An adeguate corrosion control program must be in place in order to detect corrosion defects or corrosion clusters.	Monitored data shows that the pipeline is operated according to design. Corrosion assessment carried out on a regular basis based on the monitored data		

6.2 DUCTILE FRACTURE PROPAGATION

The management of ductile fracture propagation involves both initiation control and propagation control. Initiation control focuses on preventing the initiation of fractures, while propagation control aims to limit the extent of fracture propagation once it starts. The goal of an inspection and monitoring plan is to prevent and mitigate failures by addressing fracture initiation mechanisms. Fracture initiation control is achieved through pipeline design considerations such as wall thickness, material selection, product specifications, depth of cover, and control of operating conditions. The DNV-RP-F116 flow chart for structural threats is applied to manage these aspects (Figure 2 and Table 2).

Figure 3: General flow chart for level-1 PoF evaluation of structural threats – DNV-RP F116 [2]

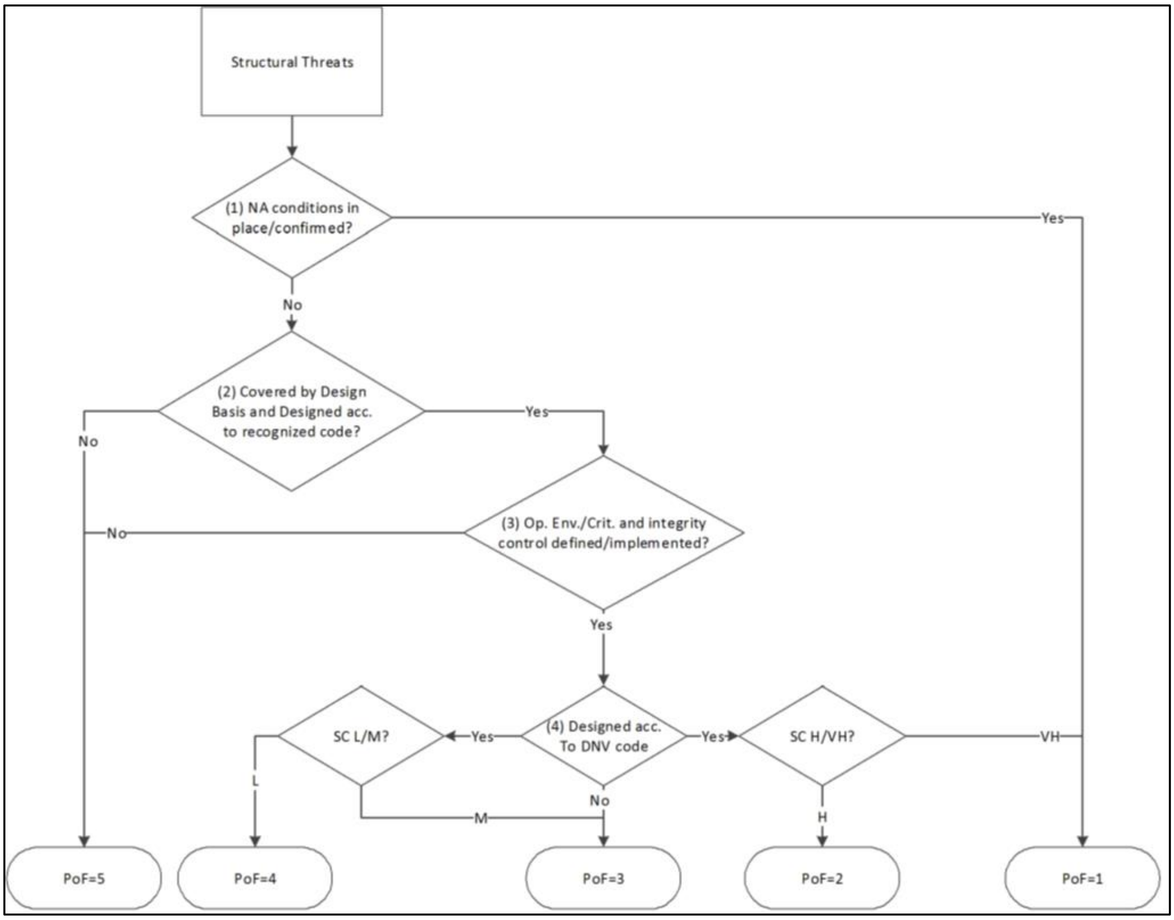


Table 2: Structural Threats assessment – DNV-RP F116 [2]

ID	Question	'YES' condition for PoF assessment (see Figure 9-3) CARBON DIOXIDE NEW DESIGN	CARBON DIOXIDE RE-PURPOSING
1	*Applicable condition in place?*	The 'non applicability' criterion is in place	
2	*Covered by design basis and designed according to recognized code?*	The design has been performed taking into account the force resisting the fracture propagation: - Pipeline geometry (wall thickness), - Material toughness and strength, - Pipeline protected by pipeline trenching and backfilling.	- Material toughness is sufficient to arrest fracture (design data, manufacturing data, mill certificates, intelligent tool for material determination). - Stream specification and operating condition changed to prevent fracture propagation - Pipeline protected by delayed pipeline trenching and backfilling.
3	*Operational envelop and integrity control defined / implemented?*	An operational envelope has been established (maximum temperature, pressure, flow-rate, trawl loads, frequencies, environmental load, impact loads, frequencies, maximum allowable span lengths, minimum cover height, etc.) and a program to check compliance is in place.	
4	*Design according to DNV codes*	Design is according to DNV codes (such a design can give a PoF category between 2 and 4 as a starting point depending on safety class and the revision of the code - for design according to other codes, a PoF category of 3 is assumed as a starting point)	For a re-purposed pipeline, the design could not be in accordance with DNV codes for CO_2 pipelines (PoF=3). 'YES' condition to be evaluated by engineering judgement.

6.3 External Coating Damage

Usually, O&G Operators Risk Analysis Methodologies include external coating damage inside the structural threats group. Therefore, the general flow chart for external corrosion provided in Figure 4 should be applied.

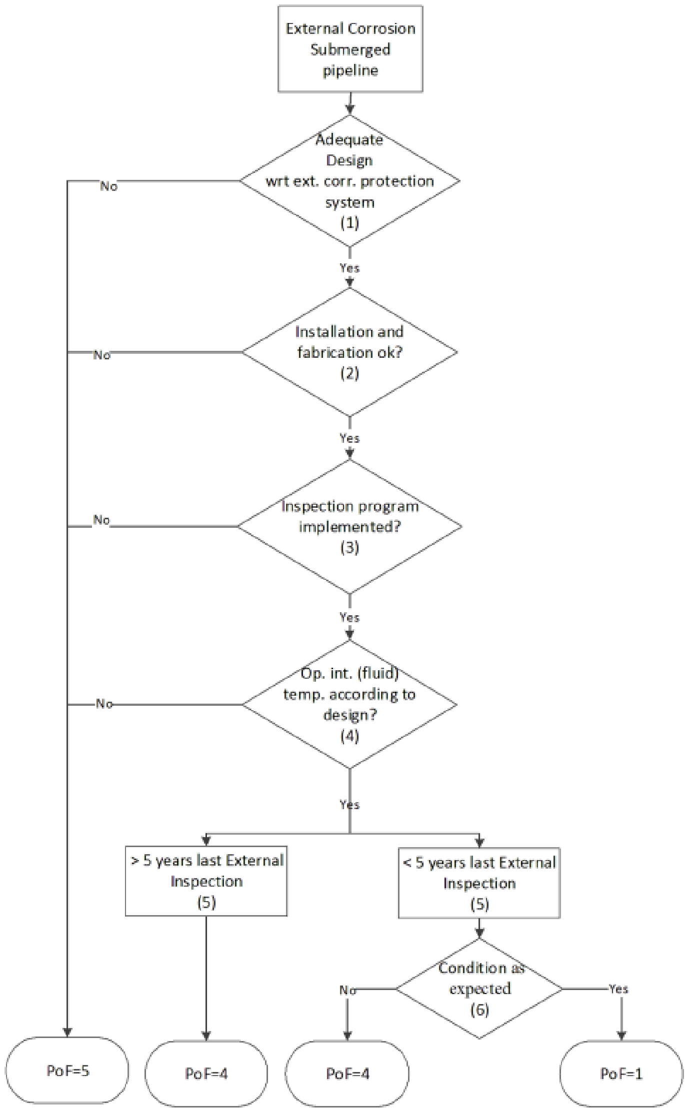
The following should be carefully evaluated both in case of new pipeline and re-purposed pipeline:

- The external coating should be qualified for low temperatures (design according to recognized codes / DNV code),
- Operation Envelope shall be established with particular attention to maximum temperature, pressure and flow-rate. A program to check compliance must be in place.

According to Ref. 12, test performed at -30° on a range of Field joint coating materials, have shown good results for 3LPE and 3LPP coatings.

Visual inspections, including cathodic protection assessments, are essential. Offshore inspections utilize ROVs with cameras and probes, while onshore inspections employ portable instruments or installed sensors.

Figure 4: General flow chart for level-1 PoF evaluation of external corrosion threats – DNV-RP F116 [2]



6.4 Non-Metallic Materials

In case of pipeline designed for CO_2 transport, the inspection and maintenance plan shall be developed by ensuring that the design of the materials is realized according to relevant codes and regulations. In particular:

- Internal coating or lining materials shall be qualified for the design conditions and for operation inside CO_2 environment.
- O-rings, seals, valve seats etc. shall be qualified to ensure the ability to resist destructive decompression, the chemical compatibility with CO_2 and other chemical components in the CO_2 stream without causing decomposing, hardening or significant negative impact on key material properties, and the functionality in the full temperature range.
- Compatibility of the applied lubricants with CO_2 shall be documented for the specified CO_2 composition and operating envelope in terms of pressure and temperature.

The re-purposing of a pipeline previous transporting hydrocarbons will most likely require the replacement of these components with components adequately designed for operation in CO_2 environment.

The following inspection and monitoring strategy should be applied:

- A proper operational and process control should be applied with the aim to avoid/control potential low temperature conditions associated with rapid pipeline depressurization leading to coating blistering and/or detachment.
- In case of low temperature condition, the decompression effects may be gradual, i.e. start as blistering and ultimately cause full detachment. In this case, an in-line inspection campaign with tools able to detect coating defects, should be adopted to detect internal lining defects and keep under control the internal coating degradation.

Table 3: IMR Plan Guide

THREAT GROUP	THREAT	Potential Initiator	Consequences	Impact of CO_2	Prevention/Mitigation	MONITORING
DFI threats	Design errors	-	- Normally covered through QA/QC during DFI - Normally occurs during installation and early operation: not part of the long term inspection program (Refs. [C-2] and [C-3])			
	Fabrication related					
	Installation related					
Corrosion/Erosion Threats	Internal corrosion	- Process fluid out of spec.	Corrosion leading to leak	- Presence of water and other impurities (corrosive ambient)	- Fluid control: level of impurities within the acceptable limits	- In-line Inspection
	External corrosion	- CP malfunctioning	Corrosion leading to leak	- CP malfunctioning due to uncontrolled depressurization leading to very low temperatures	- Process control: avoid sudden depressurizzations and or temperature monitoring	- External survey - In-line Inspection - CP Survey
	Erosion	*Same as per hydrocarbons service pipelines*				- External survey - In-line Inspection - CP Survey
	Stress corrosion cracking					
	Anode damage – CP effectiveness					
Third party threats		*Same as per hydrocarbons service pipelines*				- External survey - In-line Inspection
Structural threats	Global buckling (exposed), Global buckling (buried), End expansion, On-bottom Stability, Static Overload, Fatigue	*Same as per hydrocarbons service pipelines*				- External survey - In-line Inspection
	Ductile fracture propagation	- Excessive internal pressure - Corrosion -Third party interference	Rupture	- Decompression behaviour - Presence of impurities	- Process control: avoid sudden depressurizzations - Protective means against external interference (berms, pipeline burial, forbid areas to navigation)	- In-line Inspection (Material Properties) - In-line Inspection (Metal Loss)
	External coating damage	- External coating not qualified for low temperatures - Third Party interference	External Corrosion leading to leak	- Uncontrolled depressurization leading to very low temperatures	- Process control: avoid sudden depressurizzations - Protective means against external interference (berms, pipeline burial, forbid areas to navigation)	- Visual and Instrumental Survey
Natural hazard threats		*Same as per hydrocarbons service pipelines*				- External survey - In-line Inspection
Non-metallic Materials	Internal Coating Damage	CO_2 permeation + rapid pipeline depressurization causing temperature lowering	Internal Coating Detachment	- Decompression behaviour	- Adequate design of non-metallic internal lining - Process control: avoid sudden depressurizzations	- In-line Inspection
	Non-metallic Seals Damage	- CO_2 permeation + rapid pipeline depressurization causing temperature lowering - CO2 interaction	- Destructive Decompression - Hardening - Impact on key material properties	Interaction with CO_2 environment	- Adequate design of non-metallic seals	- External Visual Inspection
	Lubricants	Deterioration by CO_2	Deterioration		- Adequate design of lubricants	- External Visual Inspection
Incorrect Operation	Incorrect procedures	-	Covered by review/audits and training of personnel.			
	Procedures not implemented	-				
	Human errors	-				
	Internal protection system related	-				
	Interface component related	-				

7 REAL CASE APPLICATION

A real case application is presented considering the re-purposing of a 34-inch pipeline (exposed pipeline), with corrosion defects not requiring any kind of intervention for hydrocarbon transport, and with 3LPE external anticorrosion coating. The section considered is the offshore section (un-manned).

The methodology for IMR development is applied considering that the pipeline system is suitable for re-purposing to CO_2 transport and all the design data requirements have been verified for re-purposing scope to meet design requirements (see Figure 3 and Table 2).

The consequence of Failure (CoF) is evaluated according to the Level-1 Approach - Aggregated Model as proposed by Ref.2 . Ref.1 classifies carbon dioxide as Fluid Category E. A summary of the main outcomes is presented in Table 4.

Table 4: Case Study Results

THREAT GROUP	THREAT	Potential Initiator	Impact of CO_2	Prevention/Mitigation	CASE STUDY			MONITORING
					PoF	CoF[1]	RISK LEVEL (INSPECTION INTERVAL)	
Corrosion/Erosion Threats	Internal corrosion	- Process fluid out of spec.	- Presence of water and other impurities (corrosive ambient)	- Fluid control: level of impurities within the acceptable limits - Fitness for service of existing defects	3[2]	E	Risk Not Acceptable[6]	- Intelligent pigging
Structural threats	Ductile fracture propagation	- Excessive internal pressure - Corrosion -Third party interference	- Decompression behaviour - Presence of impurities	- Process control: avoid sudden depressurizzations - Protective means against external interference (berms, pipeline burial, forbid areas to navigation)	3 / 4[3]	E	Risk Not Acceptable[6]	- In-line Inspection (Material Properties and Structure) - In-line Inspection (Metal Loss)
	External coating damage	- External coating not qualified for low temperatures - Third Party interference	- Decompression behaviour	- Process control: avoid sudden depressurizzations - Protective means against external interference (berms, pipeline burial, forbid areas to navigation)	3 / 4[3]	E	Risk Not Acceptable[6]	- Visual and Instrumental Survey
Non-metallic Materials	Internal Coating Damage	CO_2 permeation + rapid pipeline depressurization causing temperature lowering	- Decompression behaviour	- Verification of existing coating properties - Process control: avoid sudden depressurizzations	2 / 3[4]	E	High (1 Year) / Risk Not Acceptable[6]	In-line inspection of the internal coating
	Non-metallic Seals Damage	- CO_2 permeation + rapid pipeline depressurization causing temperature lowering - CO2 interaction	Interaction with CO_2 environment	- Verification of material properties - Replacement with components adequately designed for operation in CO_2 environment	NA[5]	E	-	- External Visual Inspection
	Lubricants	Deterioration by CO_2		- Replacement with lubricants adequately designed for operation in CO_2 environment		E	-	- External Visual Inspection

Notes: (1) Calculated according to Level 1-Segregated Model Option (DNV-RP-F116-Table F-15).

(2) PoF=3 due to existing corrosion defects. Relaxation is allowed in case of fitness for purpose of existing defects in CO_2.

(3) Re-purposing has been evaluated and allowed according to recognized codes. In case of DNV code PoF=4 (Safety Class=Low for the specific case), otherwise PoF=3.

(4) Internal coating properties verified and re-puprosing of the pipeline accepted. Monitoring by means of intelligent tools.

(5) Not applicable for offshore section (deep water).

(6) A Level-2 assessment (or higher) is required.

8 INNOVATIVE SOLUTIONS TO IMPROVE INSPECTIONS

As illustrated in Table 3 and Table 4, the external visual inspection/instrumentation survey and the capability to constantly monitor the pipelines re-purposed for CO_2 transport play a key role in guaranteeing the safety operation and tackle the challenges inherent of this kind of application. The use of Underwater Inspection Drones (UID) to continuous monitoring of subsea assets operating in a subsea residence mode is no longer a research and development need but is already being promoted by Saipem as a game changer for IMR activities.

The possibility to have a subsea vehicle equipped with a suitable set of sensors that can merge data acquisition with Artificial Intelligence online processing allows a prompt identification of threats and immediate trigger routines to prevent asset damages. Saipem already have two UIDs in operation:

- **FlatFish** is an advanced inspection drone conceived to monitor and maintain subsea infrastructures in an innovative manner, combining the most advanced sensing and monitoring devices enabling functionalities such as underwater pipeline and riser autonomous inspection, subsea IoT data harvesting, contactless monitoring of cathodic protection systems.

- **Hydrone-R** is a subsea intervention drone able to stay subsea for a long time without vessel support, providing immediate response relying on manipulation capabilities with advanced autonomous inspection functionalities. Is the most versatile drone configuration inside Saipem Drones fleet when survey and inspection activities must be carried out in combination with light intervention works.

The combination of correct design inputs, a suitable integrity management plan with innovative solutions in the inspection field could de-risk IMR activities related to re-purposed CO2 transportation pipelines.

9 CONCLUSION

The transportation of CO_2 through pipelines (either new or re-purposed) presents unique and complex challenges that require tailored engineering solutions and specialized operational procedures. Effective management of potential threats affecting a pipeline system is crucial to ensure integrity and safety.

For a CO_2 pipeline system, the impact of the transported medium on well-known threats (internal and external corrosion) must be considered and the most appropriate actions should be taken. On the other hand, operators must be also ready to face threats not experienced in hydrocarbons service (degradation of non-metallic materials), or threats whose consequence are of bigger impact (ductile fracture propagation). These aspects must be addressed (highlighting the most critical points that deserve attention and probably dedicated analyses) during any risk review session performed with the aim to define the most appropriate monitoring and inspection plan for the specific pipeline system.

The analysis presented in this paper highlights the importance of applying correct operation procedures for the transportation of CO_2 through pipeline systems. Deviation from optimal transport conditions can indeed have a negative effect on the evolution of all threats considered, either new or known threats. Water dropout or strong temperature lowering due to sudden depressurization are some examples.

The qualification of non-metallic materials plays a crucial role in the operation of a pipeline system transporting CO_2: rapid degradation of these materials would cause significant losses in terms of cost and intervention time. However, materials being specifically certified to operate in a CO_2 environment are under testing and qualification to date: appropriate inspection and monitoring plans could prevent the failure of such components while the system is in operation.

When considering the re-purposing of a pipeline from hydrocarbons to CO_2, the availability of the steel material properties becomes essential (resistance to ductile fracture propagation). This can be obtained by reviewing available design data, manufacturing data and mill certificates. As an alternative, material determination can be achieved by specific intelligent tools capable of determining material toughness and microstructure (hard-spots). On the other hand, latest inspection data should be available in order to evaluate the internal condition of the pipeline: corrosion anomalies previously detected and accepted for hydrocarbons operation, should be carefully re-evaluated to understand as they would be expected to grow during service of a carbon dioxide pipeline. On the contrary, a first inspection campaign followed by a fitness for service assessment become indispensable.

The real case study performed for a re-purposed pipeline put into evidence the ‘non-suitability’ of the simplified approach as over-conservative results are achieved. In this case different approaches requiring in-depth analyses are required.

Continuous research and development in this field are essential to bridge existing gaps in current standards and guidelines, allowing for accurate assessment of design requirements and a more optimized threat management and for this reason Saipem joined in 2023 the CO2SafePipe JIP (Ref. 11) led by DNV that is working to narrow or close some of the gaps also identified in this paper.

As technology and practices evolve, the implementation of risk-based inspection and maintenance plans remains fundamental to addressing the specific challenges of CO_2 transportation and ensuring safe and efficient operations in the long term.

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